Sujay Kumar

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Education

- **Gurukul Kangri Deemed to be University** — Haridwar, India
  Bachelor of Technology - Electronics and Communication Engineering — *Oct 2021 - June 2025*
  Relevant Courses: *Data Structures, Programming with JAVA and Python, Object Oriented Programming*
- **Trident Public School** — Muzaffarpur, India
  Class- XII—CBSE — *Apr 2019 - Mar 2021*

Skills Summary

- **Programming Languages**: Python, Java, SQL
- **Machine Learning & AI**: Supervised Learning, Unsupervised Learning, Time Series Forecasting, Deep Learning, NLP, Feature Engineering, Model Evaluation
- **Frameworks & Libraries**: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
- **Data & Tools**: MongoDB, MySQL, Firebase, Git, Jupyter Notebook
- **Core Concepts**: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Optimization Techniques
- **Cloud & Platforms**: Linux, Windows, REST APIs, Web Technologies
- **Soft Skills**: Problem Solving, Analytical Thinking, Leadership, Time Management

Experience

- **Yathaarth Sangam Foundation (NPO)** — Remote
  Web Development Intern (Full-time) — *Nov 2024 - June 2025*
  - **Data-Driven Web Application**: Developed a scalable web application using React.js and Firebase, optimizing Firestore (NoSQL) data structures to improve data retrieval efficiency by 30%, enabling faster data-driven decision workflows.
  - **Backend Optimization**: Improved backend data handling and query performance through efficient schema design and indexing strategies, enhancing application responsiveness under dynamic data loads.
- **Aryabhatta Research Institute of Observational Sciences** — Nainital
  Project Intern (Full-time) — *Aug 2024 - Oct 2024*
  - **Data Processing System**: Built a data-centric web application using FastAPI, React, and MongoDB, optimizing data pipelines for efficient data preprocessing, transformation, and analysis and improving processing efficiency.
  - **API & Data Integration**: Designed RESTful APIs for structured data flow between frontend and backend, enabling efficient data handling, transformation, and retrieval for research workflows.

Projects

- **Financial Time Series Forecasting using Hybrid Deep Learning Model — Python, TensorFlow, Optuna, Scikit-learn**:
  - **Model Design & Feature Engineering**: Designed a hybrid CNN + LSTM model and engineered financial indicators (RSI, MACD, VWAP, ADX) for S&P 500 dataset.
  - **Optimization**: Applied hyperparameter tuning using Optuna and GPU-accelerated mixed precision training to improve model training efficiency.
  - **Evaluation**: Achieved improved directional accuracy and reduced prediction error compared to baseline models using MAE, RMSE, and R^2 metrics
- **MindTrack App — React, TypeScript, Firebase, Turso (libSQL)**: GitHub Link
  - **System Design**: Built a cross-platform application with scalable data handling and API-driven architecture, improving user retention by 35%.

Volunteer Experience

- **Web Development Lead — Jnanagni College Fest** — Haridwar, India
  Gurukul Kangri Deemed to be University — *Sept 2023 - Nov 2023*
  - **System Design & Data Handling**: Developed a scalable event management system using TypeScript and Firebase, enabling structured data handling and real-time updates for large-scale event operations.
  - **Automation & Optimization**: Built an Android-based QR verification system, reducing event management time by 30% and improving security efficiency by 45% through automated validation workflows.
- **Core Team Member — Google Developer Student Club (GKV)** — Haridwar, India
  Gurukul Kangri Deemed to be University — *Jan 2022 - Sept 2023*
  - **Collaboration & Technical Engagement**: Collaborated on technical initiatives and peer learning sessions, contributing to knowledge sharing in software development and emerging technologies.

Certifications

- **Machine Learning Specialization**: DeepLearning.AI — *Feb 2026*
- **Deep Learning Specialization**: DeepLearning.AI — *Feb 2026*
- **Natural Language Processing Specialization**: DeepLearning.AI — *Feb 2026*
- **Mathematics for Machine Learning and Data Science**: DeepLearning.AI — *Oct 2025*
- **Programming with JavaScript**: Meta — Coursera — *Feb 2024*
- **Data Structures and Algorithms (Java)**: Coding Ninjas — *Sept 2023*